

Builder: Maletso Hloane - Individual
Build Name: ClimaAware - Smart
Indoor Weather Awareness System
Location: Mopani tvet college,
Limpopo, South Africa
Category: Technomania

Solution Overview:

- Solution Overview: The solution is ClimaAware, a Smart Indoor Weather Awareness System that displays real outdoor weather conditions inside the house at the main exit door. The system uses an outdoor DHT22 sensor to detect real-time outside temperature. It then presents that information indoors through a 0.96 inch OLED display panel and two simple LEDs. If it is cold below 18°C, the RED LED turns ON and the display shows the temperature with a jersey icon, indicating warm clothes are needed. If it is hot above 30°C, the RED LED also turns ON with a sun icon warning to avoid sun. If temperature is normal between 18-30°C, the GREEN LED turns ON indicating it is safe to play. This corrects the body's false feeling and uses a simple rule for children: GREEN = Go Play, RED = Wait for Parents. No voice, no RGB, only clear Green and Red.

Everything is in that kit below

- The kit is from takealot
- <https://www.takealot.com/basic-starter-kit-for-esp32-32s-wifi-iot-development-board-with/PLID100195334>
- Total cost = R 490

Marketability & Validation Section:

How will i market this prototype:

ClimaAware is marketable worldwide because the problem is global: every house in the world is warmer inside than outside in winter, children feel warm after a bath and run outside without a jersey and get sick. This happens in Africa, Europe, USA, Asia. Validated by 2 mothers in Namakgale, Limpopo and supported by WHO data on children catching colds from improper cold exposure.

How people around the world will use it:

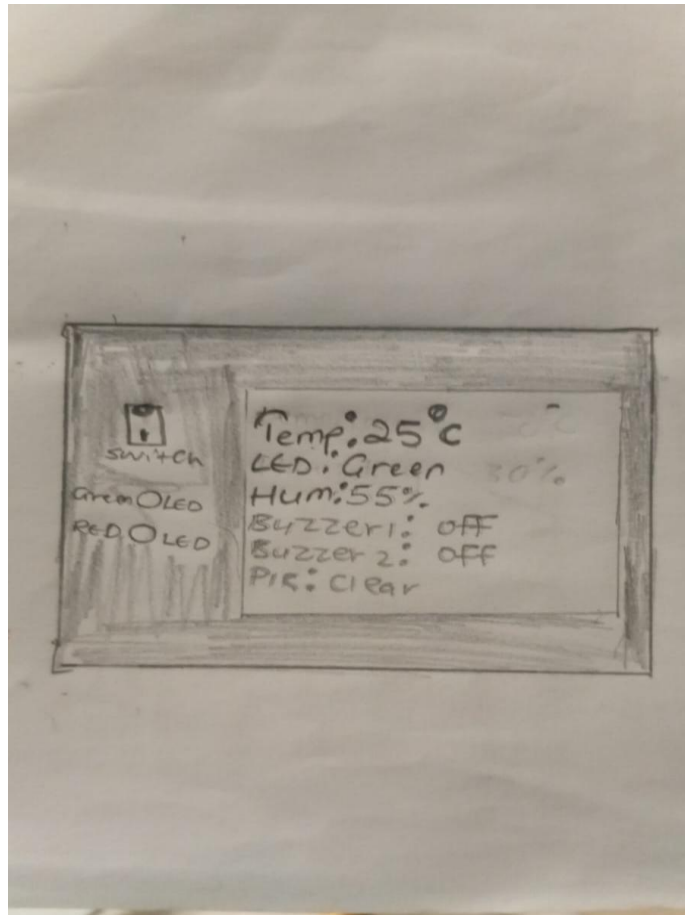
- Device is fixed at the main exit door. No voice, no RGB, only Green and Red LEDs that any child in any country understands without language. GREEN ON = Safe to go outside. RED ON = STOP, need warm clothes. OLED shows temperature in °C and °F with universal icons: Jersey for cold below 18°C and Sun for hot above 30°C. Works without WiFi, without internet, without subscription, just plug into 5V phone charger.

Problem exist

- To validate that this problem is real and not just my own experience, I conducted informal market research in my community in Tzaneen, Limpopo. I interviewed 2 mothers of young children:
- Mother 1: Said her child frequently catches a cold in winter after going outside without warm clothes after bathing.
- Mother 2 Said she checks her phone for weather, but her children do not, so they ignore her warnings.

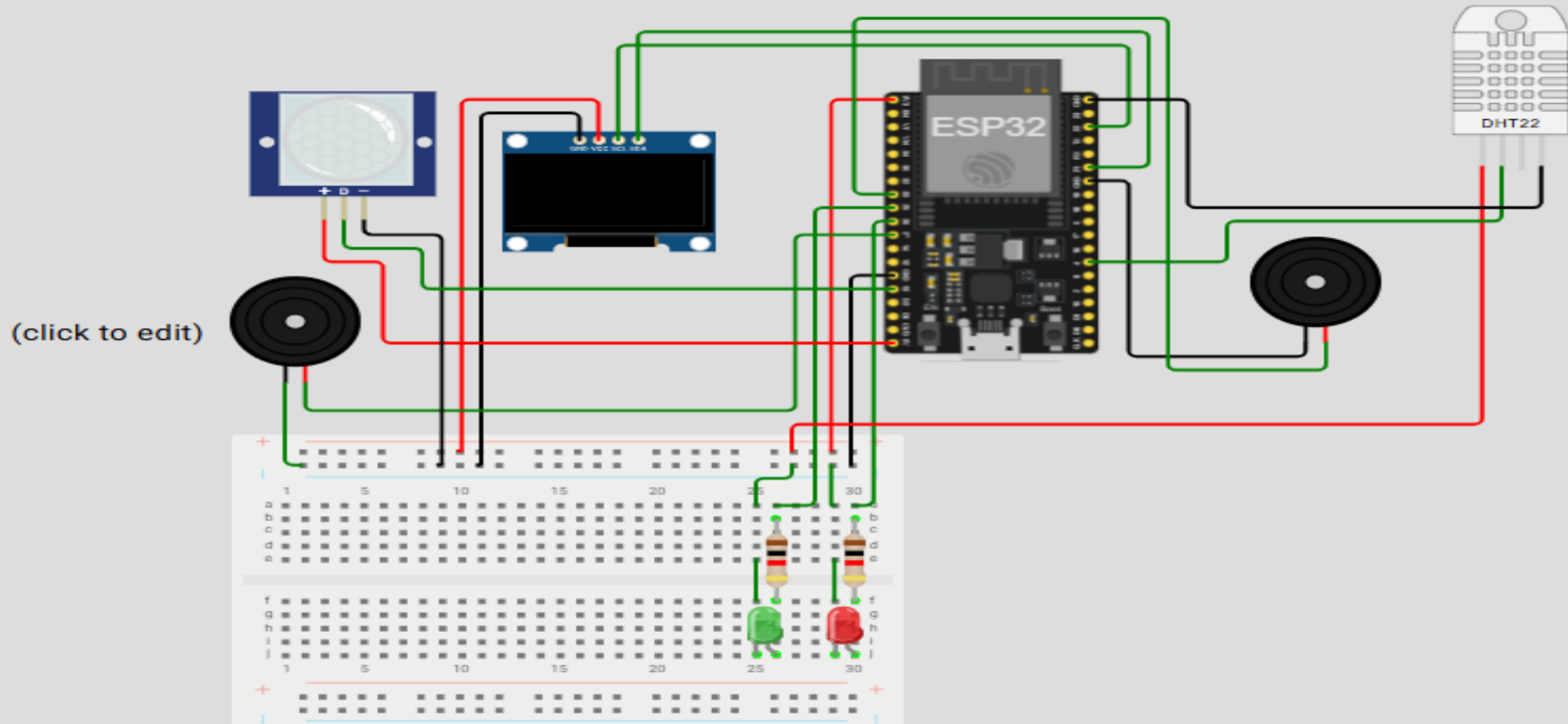
confirmed they would want a simple device at the door that tells the child what to do, like "Wear Jersey" or "Avoid Sun". This validates that the problem exists in real homes and there is a need for a low-cost, visual, child-friendly solution at the point of exit.

Technical implementation Section



- • ClimaAware is a smart child safety device mounted on the side of the door that monitors outdoor weather conditions to prevent children from going outside in dangerous climates. It uses a DHT22 sensor to measure temperature and humidity, and it displays the live readings on an OLED screen along with the status of the green LED for safe conditions, red LED for unsafe conditions, Buzzer 1 that beeps slowly when the temperature is too hot, too cold, or humidity is too high, Buzzer 2 that beeps fast fast when the PIR motion sensor detects a child trying to open the door during unsafe weather, and the PIR status itself. In your picture, it shows a safe condition at 25°C and 55% humidity with the green LED on, both buzzers off, and PIR clear, meaning a child can safely go outside, while if the temperature drops below 18°C or rises above 30°C the system automatically switches to the red LED and activates the buzzers to warn and stop the child

Electronics



In this file below is code and design(wiring)

I used wokwi to make a design (wiring)



project (4).zip